Large-Scale Anonymized Text-based Disability Discourse Dataset

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ABSTRACT

The involvement of individuals with disabilities in online discussions related to disability and accessibility is a critical area of study. While previous research has qualitatively examined the participation of individuals with disabilities on social media platforms, large-scale analysis of social media content by people with disabilities has been an underexplored area. This paper presents a pioneering large-scale study of disability communities on Reddit. We developed an anonymized text-based dataset that consists of 1.5 million comments posted on three subreddits: r/disability, r/Blind, and r/ADHD. Using topic modeling, we analyzed the dataset and identified eight highly-coherent common categories and their associated keywords across the three subreddits. We contribute an *Anonymized Disability Discourse Reddit Corpus (ADDReC)* of 1.5 million comments that feature eight disability discourse categories.

CCS CONCEPTS

• **Human-centered computing → Empirical studies in collaborative and social computing**.

KEYWORDS

disability, discourse, dataset, blind, ADHD, Reddit

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1 INTRODUCTION

Accessibility researchers often struggle to engage people with disabilities in their projects due to various factors, such as people with disabilities' unwillingness or inability to participate [13, 15, 16, 24]. Traditional methods of data collection, such as surveys and interviews, are also prone to biases and limitations [8, 20, 27]. Fortunately, social media provides an avenue for individuals with disabilities to engage in discussions about topics that are relevant and interesting to them [18, 25, 26, 28]. Exploring these online communities can help alleviate the burden of survey requests and offer preliminary insights into the accessibility desires and needs of people with disabilities. Previous research has primarily focused on qualitatively examining social media content created by people with disabilities and was often limited to specific disability groups, such as blind individuals [9, 11, 23]. In complementing prior work, our study aims to create a text-based dataset of social media content. This dataset will allow researchers to gain a deeper understanding of the lived experiences of individuals with disabilities and to identify accessibility concerns raised by them.

We collected data from Reddit because of the high-quality comments produced on Reddit, as evidenced by previous research [19] where different subreddits were analyzed such as parenting [2, 10], science communication [3], and mental health [6]. To ensure inclusivity and representativeness within the disability community, we collected data from three distinct subreddits: a general disability subreddit[1], a subreddit focused on physical disabilities (specifically blindness)[2], and a subreddit dedicated to neurodiversity (specifically ADHD)[3]. We compiled a dataset of 1,722,769 comments posted on these subreddits and anonymized it. We applied the Latent Dirichlet Allocation (LDA) model [4] to analyze the data, as LDA is a widely used technique for topic modeling in text data and social media analysis [1, 7, 14, 17, 21]. Our analysis revealed eight coherent categories and their keywords. These categories represent the common topics discussed on the three subreddits. This paper presents two **contributions**. First, we developed an **A**nonymized **D**isability **D**iscourse **Re**ddit **C**orpus (ADDReC) [4]. Second, we provided an overview of the topics present in the corpus.

Ethical Considerations. Our research was classified as non-human-subject research by our Institutional Review Board, as we did not engage with participants. We obtained permission from the moderators of the r/ADHD subreddit in accordance with their community rules. We used publicly available data from Reddit, which was anonymized to protect privacy, and paraphrased reported quotes.

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[1]https://reddit.com/r/disability
[2]https://reddit.com/r/Blind
[3]https://reddit.com/r/adhd
[4]https://github.com/thekindlab/addrec

2 METHOD

2.1 Data Source

We collected the data for our study in April 2021. We searched for active subreddits where the main topic of the forum is disability. Our criteria for an "active subreddit" included a minimum of 1000 members and recent posts that were not auto-generated. We ran a search query on Google with the keywords "subreddit" and "disability". The largest subreddit on disability we found that satisfied our criteria was r/disability with 25.6K members. This online community is open to conversations related to any type of disability. On the r/disability page, Reddit showed related communities. The largest subreddit focused on blindness, an example of physical disabilities, is r/Blind with 12.8K members. We found r/ADHD to be the largest neurodiversity-focused community with more than 1.2 million members.

We used Google's BigQuery [5] to download all the publicly available comments from r/disability, r/Blind, and r/ADHD in the period from January 1st, 2015 to December 31st, 2019. We included only comments with upvote scores of at least 1 to ensure that the selected comments represented typical accepted community discourse. This approach excluded comments that were negatively rated. The corpus includes users' comments, timestamps, and karma scores. The corpus initially consisted of a total of 1,722,769 comments, with 81,002 comments from r/disability, 43,026 comments from r/Blind, and 1,598,741 comments from r/ADHD. To prepare the data for analysis, we cleaned the data using two steps. First, we removed short comments that had fewer than 35 characters, which contained messages such as "Thank you". By manual validation of a sample of 900 comments, we found that these short comments were irrelevant to our study. Second, we removed comments that appear as "[deleted]" or "[removed]". These comments were deleted by their author or by moderators. After the cleaning phase, the dataset contained 1,526,980 comments.

2.2 Data Anonymization

To preserve anonymity in the comments, we employed Microsoft Presidio Anonymizer, an open-source software that utilizes various techniques such as named entity recognition, regular expressions, checksum validation, and context word analysis to identify and anonymize personally identifiable information in the dataset. To ensure comprehensive anonymization, we used the full list of Presidio-supported entities[6], including names, locations (zip codes and public named locations), phone numbers, email addresses, URLs, social security numbers, and financial information such as credit cards, bank codes, and bitcoin wallets. Additionally, we added a custom rule to mask the occurrences of usernames. The rule was manually defined to catch '/u/username/' with the regular expression /r'/u/([a-zA-Z0-9_]*)/ b'/ to capture any string that starts with '/u/' and is followed by any combination of alphanumeric characters or underscores. We imported all CSV files from the dataset into Pandas DataFrames and applied our anonymization script to each DataFrame.

[5] https://cloud.google.com/bigquery

[6] https://microsoft.github.io/presidio/supported_entities/

2.3 Topic Modeling

We utilized the Latent Dirichlet Allocation (LDA) model [12] to identify the discourse categories across the three subreddits. LDA considers each comment as a document and considers the subreddits a corpus. Before the application of the LDA model, we pre-processed all comments in three steps: tokenization, lemmatization, and stopword removal. Tokenization involves breaking down a stream of textual data into meaningful elements, such as words, resulting in a concise list of tokens. Lemmatization reduces each word to its base or root form, which helps to extract the intended meaning of the keywords. Lemmatization allows for the morphological analysis of words, enabling topic modeling techniques to understand the contextual form of each word and cluster it accordingly. Stopwords are commonly used words, such as prepositions and conjunctions, that carry little informational value and hence were removed.

The LDA model generated abstract categories of keywords and assigned coherence scores to measure their semantic similarity. Coherence scores were used to differentiate meaningful categories from statistical artifacts [22]. To ensure meaningful categories, we iteratively ran the LDA model ten times on the corpus, fine-tuning the model until obtaining high-coherence categories [5]. To ensure the reliability and validity of the LDA model's output, we sampled 100 comments for each category and manually reviewed them. Finally, we applied an automated process to determine the prevalence of the LDA categories in the subreddits. Every comment was checked for the presence of keywords associated with specific LDA categories. A higher frequency of category keywords in a comment lead to a higher score for that category. Once all the keywords were scored for a comment, the category with the highest score was determined as the most relevant based on the keywords present in the comment.

3 RESULTS AND DISCUSSION

3.1 Disability Discourse Categories and their Associated Keywords

The LDA analysis produced ten categories. Two categories had low coherence scores and we were unable to assign clear labels to them. For example, a category with a coherence score of -11.18 contained words such as "interviewer", "app", "need", and "hadley". We excluded these low coherent categories. Our analysis revealed eight highly coherent categories and their associated words: Accessibility, Asking for Advice, Daily Activities, Disability, Medical Information, Social Services, Social Situations, and Work and School (Table 1). The most discussed topic across all three subreddits was found to be "**Social Situations**", comprising 35.5%, 33.7%, and 30.1% of posts in r/disability, r/Blind, and r/ADHD, respectively (refer to Figure 1). Examples of Social Situations comments from:

r/disability: "*People think that wheelchair users are paralyzed. Try ignoring them or explain that different disabilities require different adaptive equipment. People can be shockingly cruel.*"

r/Blind: "*I have been attending events with new people lately and I've noticed that introducing my vision impairment strengthens connections and fosters empathy.*"

Table 1: Summary of the LDA topics with high coherence in the subreddits disability, blind, and ADHD, and their associated words. The coherence score for each topic is represented by r.

Category	Associated Words in r/disability	Associated Words in r/Blind	Associated Words in r/ADHD
Accessibility	wheelchair, need, time, accessible, use, go, stall, room, work, bathroom (r=-2.57)	use, screen, voiceover, app, using, mode, people, reader, text, blind (r=-1.78)	help, book, brain, focus, music, find, use, watch, mind, sound (r=-2.61)
Asking for Advice	reddit, disability, message, comment, removed, question, automatically, compose, concern, submission (r=-1.19)	reddit, comment, blind, thread, bot, totesmessenger, someone, game, rule, place (r=-1.83)	question, comment, rule, wiki, reddit, anyone, reply, adhd, doe, message (r=-1.13)
Daily Activities	day, shoe, video, work, year, look, time, pant, wear, go (r=-2.61)	game, play, see, question, help, sound, thing, know, blind, fun (r=-1.89)	day, sleep, time, go, eat, hour, exercise, food, bed, night (r=-2.41)
Disability	chair, use, wheelchair, need, pain, help, know, doctor, issue, look (r=-1.96)	blind, people, vision, thing, know, think, use, see, impaired, visual (r=-1.88)	adhd, people, symptom, disorder, anxiety, child, depression, problem, believe, treatment, (r=-2.4)
Medical Information	doctor, medical, treatment, disability, condition, work, record, year, case, need (r=-1.61)	hospital, know, vision, thing, people, use, go, eye, near, blind (r=-1.78)	med, medication, adderall, effect, mg, doctor, taking, vyvanse, dose, day (r=-1.79)
Social Services	ssi[7], work, benefit, ssdi[8], income, month, disability, state, help, know (r=-1.55)	blind, people, time, work, need, technology, state, survey, thing, training (r=-2.91)	NA
Social Situations	people, think, disability, thing, disabled, way, need, know, say, even (r=-1.45)	blind, braille, people, use, know, might, completely, others, question (r=-2.17)	people, thing, feel, know, think, time, way, want, even, adhd (r=-1.39)
Work and School	people, disability, disabled, person, work, job, think, time, thing, able (r=-1.47)	audio, blind, work, people, braille, thing, know, description, company, use (r=-2.09)	year, school, time, job, life, college, diagnosed, class, adhd, high (r=-1.83)

r/ADHD: "*My boyfriend has been supporting me emotionally for the past 8 years. Over time, he has developed a deep understanding of me and knows that I never intend to be unkind to him.*"

The second most frequent discourse category on the three subreddits was "**Disability**". It ranked second on r/disability with respect to number of posts, forming 17.34% of the posts. It ranked third, with a small margin, on r/Blind and r/ADHD, comprising 18.52% and 21.87% of posts, respectively. Examples of Disability comments from

r/disability: " *I prefer physiotypical as an analogy to neurotypical. I have a significant physical disability, but can do things with my body that some of my friends without disabilities can't, so why are they 'able-bodied' while I'm not?*"

r/Blind: "*As someone with RP, a condition that affects low-light and night vision, I found a mobility cane necessary.*"

r/ADHD: "*I'm considering getting a nightguard. If it helps with the anxiety and ADHD-related clenching, I'm definitely giving it a try!*"

We observed differences in the frequency of other categories across the three subreddits. For instance, the "**Accessibility**" category was the second most frequent category in r/Blind, with 20.8% of the posts related to accessibility, compared to 8.9% and 3.4% of posts in r/disability and r/ADHD, respectively. Examples of Accessibility comments from

r/disability: "*The designated wheelchair-accessible space is often misused by tenants, who place large plants, extra restaurant tables, or other items in that area.*"

r/Blind: "*Screen readers ignore CSS. Thus, text can be hidden from non-screen reader users by being placed offscreen or made invisible using CSS, but screen readers will still read it.*"

r/ADHD: "*No apps work for me because I'll just exit the app or turn off my phone completely.*"

The "Medical Information" category was the second most discussed on r/ADHD, with 22.4% of the posts related to medical information, while only 10.9% and 10.3% of posts on r/disability and r/Blind were medical-related. Another discourse category of interest to users on r/disability was "Social Services", which ranked third in frequency, with 12.1% of posts related to social services, compared to 1.7% and 1.3% of posts on r/Blind and r/ADHD, respectively.

In summary, our analysis revealed that the most commonly discussed topic across all three subreddits was "Social Situations", followed by "Disability". Furthermore, each subreddit had its own most frequently discussed topics.

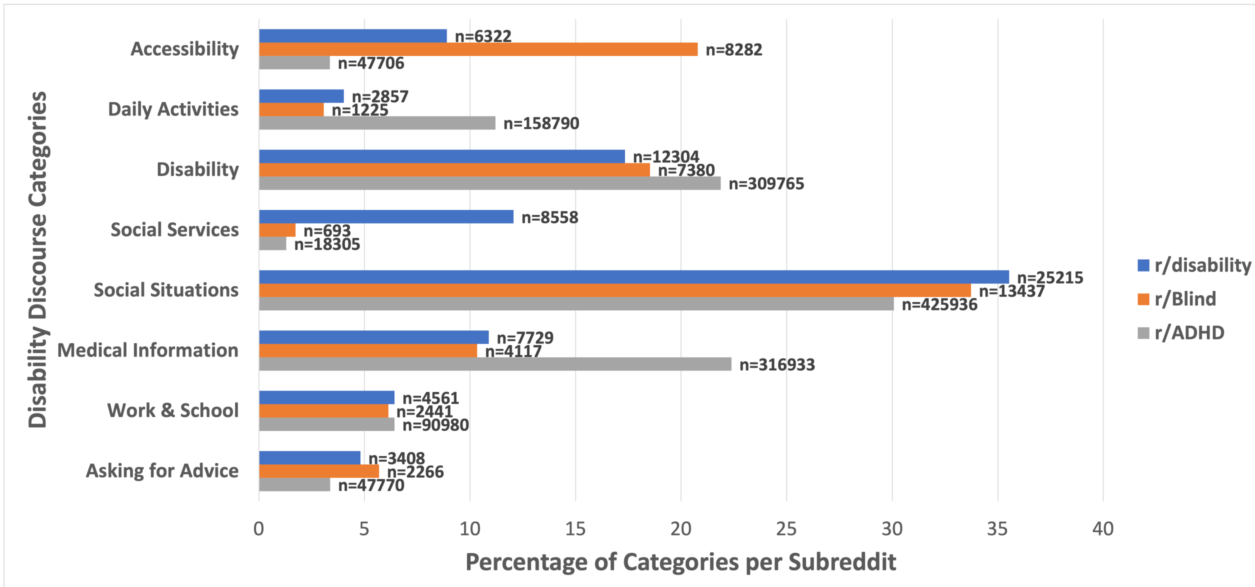

Figure 1: Distribution of comments per disability discourse category in the three subreddits.

3.2 Disability Discourse Dataset Potential Applications

Our research yielded valuable results in terms of the development a large-scale anonymized text-based dataset, ADDReC, and the analysis of discourse topics. These outcomes have the potential to facilitate comprehensive quantitative and qualitative analysis of the dataset. ADDReC provides an opportunity for a holistic analysis, enabling a deeper understanding of the intersectionality between blindness, ADHD, and other disability experiences present in the data. Such an approach can contribute to the development of inclusive technologies, environments, and services. Additionally, a holistic analysis can reveal overlapping themes, challenges, and coping mechanisms that are relevant to individuals with different disabilities. Alternatively, ADDReC can be analyzed with a focus on a specific disability, allowing for a more in-depth exploration of that particular disability and informing the development of accessible technologies tailored to the needs of blind people and people with ADHD. In constructing our dataset, we included Reddit comments along with their timestamps and karma scores. Preserving the timestamp information allows for future investigations into the evolution of disability discourse over time. Additionally, the karma scores can support future research in identifying the most popular comments within the dataset.

Limitations We recognize that potential biases might exist in the dataset, including selection bias given that Reddit users may not be representative of the broader population, and moderator bias as subreddit moderators influence the content and direction of discussions. These considerations should be taken into account by future researchers when interpreting the findings. The anonymization process using Presidio inadvertently masked some medication names due to their resemblance to people or place names, suggesting the need for a more sophisticated anonymization technique to preserve medical information.

4 CONCLUSION

We employed mixed methods to identify discourse categories in ADDReC, the dataset of 1.5 million comments which has potential value for future research in this area. Future research could expand its scope by exploring a broader range of social media platforms and including other groups of individuals with disabilities.

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